Assignment

Task 2:

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CPU SCHEDULING RESULTS
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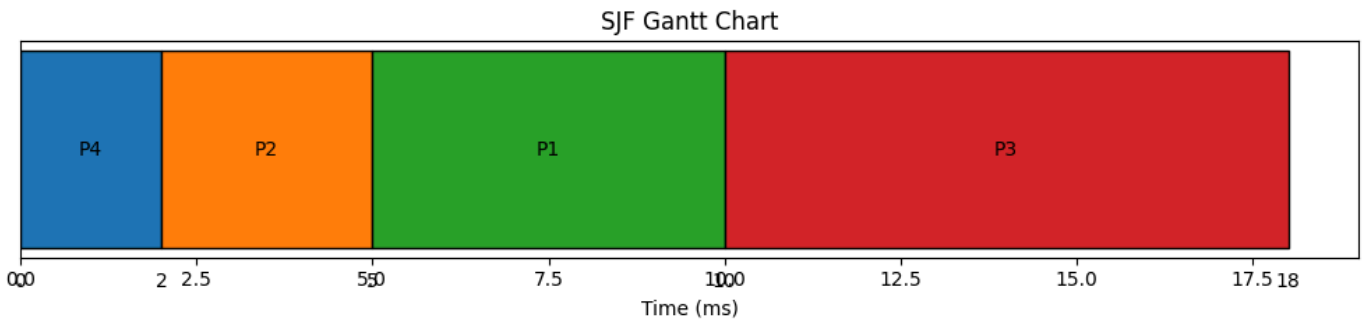
FCFS
P1 -> WT = 0 ms, TAT = 5 ms
P2 -> WT = 5 ms, TAT = 8 ms
P3 -> WT = 8 ms, TAT = 16 ms
P4 -> WT = 16 ms, TAT = 18 ms
Average WT = 7.25 ms

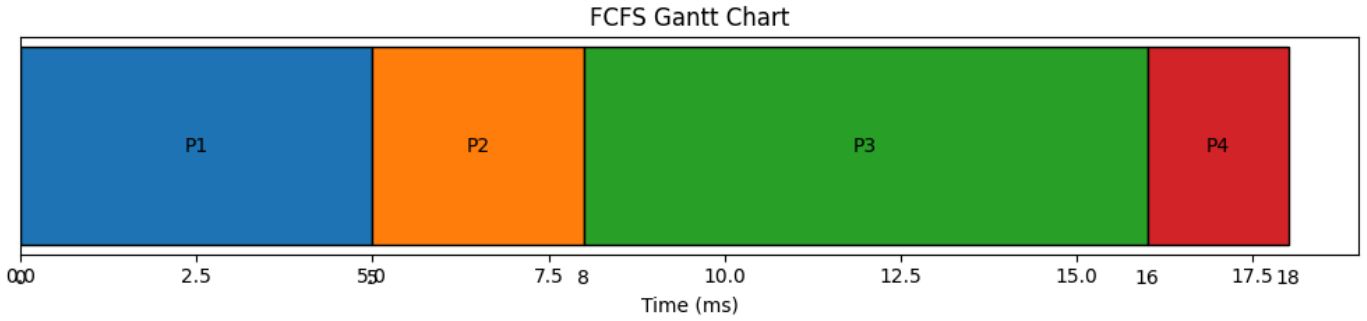
SJF
P4 -> WT = 0 ms, TAT = 2 ms
P2 -> WT = 2 ms, TAT = 5 ms
P1 -> WT = 5 ms, TAT = 10 ms
P3 -> WT = 10 ms, TAT = 18 ms
Average WT = 4.25 ms

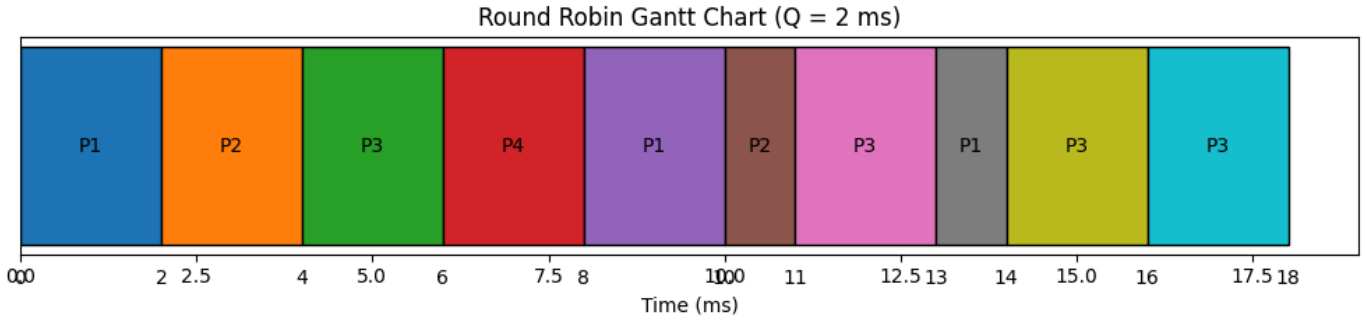
ROUND ROBIN
Quantum = 2 ms
P1 -> WT = 9 ms, TAT = 14 ms
P2 -> WT = 8 ms, TAT = 11 ms
P3 -> WT = 10 ms, TAT = 18 ms
P4 -> WT = 6 ms, TAT = 8 ms
Average WT = 8.25 ms

PRIORITY
Lower number = Higher Priority
P2 -> Priority = 1, WT = 0 ms, TAT = 3 ms
P4 -> Priority = 2, WT = 3 ms, TAT = 5 ms
P1 -> Priority = 3, WT = 5 ms, TAT = 10 ms
P3 -> Priority = 4, WT = 10 ms, TAT = 18 ms
Average WT = 4.50 ms


================================================================
AVERAGE WAITING TIME COMPARISON
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FCFS      : 7.25 ms
SJF       : 4.25 ms
RR        : 8.25 ms
Priority  : 4.50 ms
```







Task 3:

In DNN training, a scheduling decision can cause some workers or GPUs to receive slower or more demanding tasks than others, creating “stragglers” that finish later than the rest. Since distributed training often has to wait for all workers to complete their work before synchronizing, even one slow worker can delay the entire training iteration and increase the total training time.